

MLOPS IN PRODUCTION · ISSUE 02 OF 08

Canary releases and rollback for models.

If rollback needs a retrain, you do not have a rollback.

02

AR

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Swipe →

Shadow, canary or A/B?

STRATEGY	USER IMPACT	ANSWERS
Shadow	None	Does it behave sanely?
Canary	Small %	Is it safe at scale?
A/B test	Split	Is it better for the business?



Progressive rollout

- 1 Shadow
- 2 5% canary
- 3 25%
- 4 50%
- 5 100%

Hold each step long enough to see delayed labels, not just latency.

Guardrail metrics

01

Error rate

02

p95 latency

03

Prediction distribution
shift

04

Fallback rate

05

Business KPI proxy

06

Cost per 1k predictions

Rollback that works

01 Automatic on breach

Guardrail crossed, traffic shifts back.

02 Previous version warm

Loaded and ready, not cold-started.

03 Schema compatible

Both versions accept the same features.

04 One action

An alias flip or traffic weight change.

Canary mistakes

DO

- ✓ Compare canary vs control, same time window
- ✓ Watch prediction distribution
- ✓ Rehearse rollback

AVOID

- ✗ Judge on latency alone
- ✗ Roll out on Friday evening
- ✗ Change features and model together

KEY TAKEAWAY

Models fail silently. Canary on behaviour, not just errors.

- Shadow, then progressive canary
- Guardrails include prediction shift
- Rollback is one action

YOUR TURN

What metric would make you roll back a model that is still returning 200 OK?

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Next in the series: #03 Data drift vs concept drift

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